INCH-POUND
MIL-R-23648/7B
4 April 1995
SUPERSEDING
MIL-T-23648/7A
15 July 1988

MILITARY SPECIFICATION

RESISTOR, THERMAL (THERMISTOR), INSULATED NEGATIVE TEMPERATURE COEFFICIENT STYLE RTH18

THIS SPECIFICATION IS INACTIVE FOR NEW DESIGN AFTER THE DATE OF THIS SPECIFICATION (SEE 6.3).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RTH18 thermistors. This style is available in resistance ratios A, B, and C. The only terminal available is S type. Resistance tolerance versus temperature characteristics F, G, J, and K are applicable through the maximum temperature of 275°C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATION

MILITARY

MIL-R-23648 - Resistor, Thermal, (Thermistor), Insulated, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbook are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:30Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

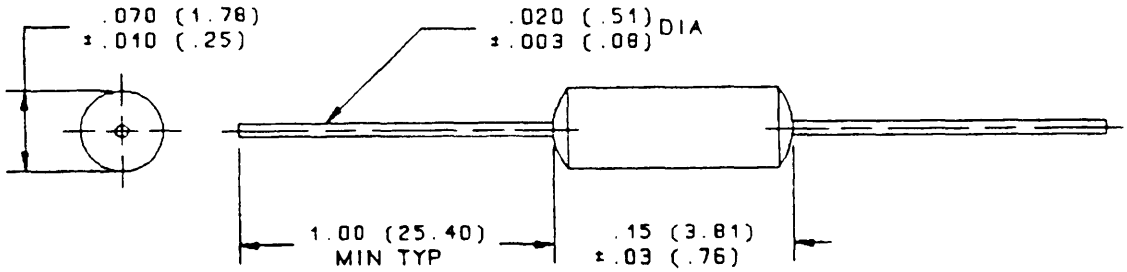
3.1 Requirements. Requirements shall be in accordance with MIL-R-23648, and as specified herein.

3.2 Design and construction. The thermistors shall be of the design, construction, and physical dimensions specified in figure 1.

3.3 Thermal time constant. The thermal time constant shall be 12 seconds maximum.

3.4 Dissipation constant. The dissipation constant shall be 2.0 milliwatts per degrees Celsius minimum.

3.5 Power rating. The thermistor shall be capable of dissipating a maximum power of 0.5 watt at 25°C. Thermistor shall be derated in accordance with figure 2.



NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.

FIGURE 1. Style RTH18 thermistor.

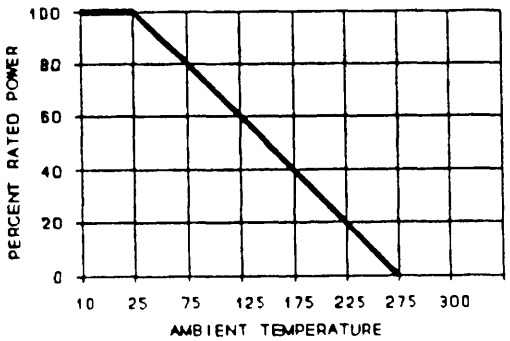


FIGURE 2. Derating curve for high ambient temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:30Z
Check the source to verify that this is the current version before use.

3.6 Resistance.

3.6.1 Resistance at 25°C. Standard values shall be as specified in MIL-R-23648 except that the 5 percent decade values shall also be used for resistance tolerances F and G. Minimum and maximum resistance values shall be in accordance with table I.

TABLE I. Minimum and maximum resistance values (at 25°C) for each available resistance ratio.

Ratio 19.8 (A)	Ratio 29.4 (B)	Ratio 48.7 (C)
(ohms) 2.2k min. 10k max.	(ohms) 22k min. 100k max.	(ohms) 220k min. 1M max.

3.6.2 Resistance at temperatures other than 25°C. Resistance at temperatures other than 25°C are determined by multiplying the zero-power resistance at 25°C by the factors shown in table II for the applicable resistance ratio.

TABLE II. Resistance temperature characteristic factors.

Temperature	Ratio 19.8 (A)	Ratio 29.4 (B)	Ratio 48.7 (C)
-55	54.9	100	- - -
-15	5.77	7.38	8.80
0	2.85	3.27	3.66
25	1.00	1.00	1.00
50	.405	.360	.320
75	.184	.148	.116
100	.0923	.0675	.047
125	.0503	.0340	.0205
200	.0119	.00660	.0029
275	.004435	.00221	.00071

3.7 Short time overload. The maximum allowable change in zero-power resistance as the result of the short time Overload test shall be ±2 percent.

3.8 Low temperature storage. The maximum allowable change in zero-power resistance as the result of the low temperature storage test shall be ±2 percent.

3.9 High temperature storage. The maximum allowable change in zero-power resistance as the result of the high temperature storage test shall be ±1 percent.

3.10 Terminal strength. When tested in accordance with 4.3, the maximum allowable change in zero-power resistance shall be ±1 percent.

3.11 Thermal shock. The maximum allowable change in zero-power resistance as the result of the thermal shock test shall be ±2 percent.

3.12 Resistance to soldering heat. The maximum allowable change in zero-power resistance as the result of the resistance to soldering heat test shall be ±1 percent.

3.13 Moisture resistance. The maximum allowable change in zero-power resistance as the result of the Moisture resistance test shall be ±5 percent.

3.14 Life. The maximum allowable change in zero-power resistance as the result of the life test shall be ±5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:30Z
Check the source to verify that this is the current version before use.

3.15 High temperature exposure. The maximum allowable change in zero-power resistance as the result of the high temperature exposure test shall be ±1 percent after 100 hours, and ±2 percent after 1,000 hours.

3.16 Vibration , high frequency. The maximum allowable change in zero-power resistance as the result of the vibration test shall be ±2 percent.

3.17 Shock, specified pulse. The maximum allowable change in zero-power resistance as the result of the shock test shall be ±2 percent.

3.18 Immersion. The maximum allowable change in zero-power resistance as the result of the immersion test shall be ±3 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification inspection. Qualification inspection is not applicable to this document.

4.2 Sampling and inspection. The sampling, acceptance inspection, and test procedures specified in table IV and V of MIL-R-23648 are applicable as sampling, inspection, and test procedures of this detail specification.

4.3 Terminal strength.

4.3.1 Direct load. Direct load shall be applied gradually until the reaches 4.5 pounds.

4.4 Solderability. The solderability test is applicable to this specification.

4.5 Resistance to soldering heat. The resistance to soldering heat is applicable to this specification.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-23648.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-23648 are applicable.

6.2 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5905-1401-04)

Source: https://assist.dla.mil -- Downloaded: 20261O-19T17:30Z
Check the source to verify that this is the current version before use.